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Combination of retagliptin and henagliflozin as add-on therapy to metformin in patients with type 2 diabetes inadequately controlled with metformin: A multicentre, randomized, double-blind, active-controlled, phase 3 trial

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Abstract

Aim: This study assessed the efficacy and safety of co-administering retagliptin and henagliflozin versus individual agents at corresponding doses in patients with type 2 diabetes mellitus who were inadequately controlled with metformin.

Methods: This multicentre, phase 3 trial consisted of a 24-week, randomized, double-blind, active-controlled period. Patients with glycated haemoglobin (HbA1c) levels between 7.5% and 10.5% were randomized to receive once-daily retagliptin 100 mg (R100; n = 155), henagliflozin 5 mg (H5; n = 156), henagliflozin 10 mg (H10; n = 156), co-administered R100/H5 (n = 155), or R100/H10 (n = 156). The primary endpoint was the change in HbA1c from baseline to week 24.

Yao Wang and Chengxia Jiang contributed equally to this work.

Results: Based on the primary estimand, the least squares mean reductions in HbA1c at week 24 were significantly greater in the R100/H5 (−1.51%) and R100/H10 (−1.54%) groups compared with those receiving the corresponding doses of individual agents (−0.98% for R100, −0.86% for H5 and −0.95% for H10, respectively; $p < .0001$ for all pairwise comparisons). Achievement of HbA1c <7.0% at week 24 was observed in 27.1% of patients in the R100 group, 21.2% in the H5 group, 24.4% in the H10 group, 57.4% in the R100/H5 group and 56.4% in the R100/H10 group. Reductions in fasting plasma glucose and 2-h postprandial glucose were also more pronounced in the co-administration groups compared with the individual agents at corresponding doses. Decreases in body weight and systolic blood pressure were greater in the groups containing henagliflozin than in the R100 group. The incidence rates of adverse events were similar across all treatment groups, with no reported episodes of severe hypoglycaemia.

Conclusions: For patients with type 2 diabetes mellitus inadequately controlled by metformin monotherapy, the co-administration of retagliptin and henagliflozin yielded more effective glycaemic control through 24 weeks compared with the individual agents at their corresponding doses.

KEYWORDS

clinical trial, dipeptidyl peptidase-4 inhibitor, metformin, sodium-glucose cotransporter 2 inhibitor, type 2 diabetes mellitus

1 | INTRODUCTION

Many patients with type 2 diabetes mellitus (T2DM) experience a decline in glycaemic control because of the disease's progressive nature, often necessitating additional therapeutic interventions. Current guidelines recommend adding a single anti-hyperglycaemic agent to the treatment regimen for patients whose glycaemic levels remain suboptimal with metformin monotherapy.[1] However, this conventional dual therapy strategy may not fully meet the needs of certain patient populations, particularly those with relatively high baseline glycated haemoglobin (HbA1c) values. Moreover, patients initially benefiting from dual therapy may encounter a deterioration in glycaemic control over time, prompting the need for further adjustments to their therapeutic approach.[2,3] In certain scenarios, clinicians may opt to enhance the treatment regimen by incorporating a combination of two oral antidiabetic drugs that exhibit complementary mechanisms of action and favourable safety profiles, without pharmacokinetic interactions.[4,5] Specifically, the concurrent use of sodium-glucose cotransporter 2 inhibitors (SGLT2is) and dipeptidyl peptidase-4 inhibitors (DPP-4is) presents a viable strategy.[3,6] This combination therapy, when used alongside metformin, targets multiple pathways, providing a synergistic effect that results in improved glycaemic control while maintaining a low risk of hypoglycaemia; in addition, it confers extra health benefits, including reductions in blood pressure (BP) and body weight, as well as cardiovascular and renal protection, increasingly supported by evidence from recent studies on SGLT2is.[7–10] This strategy might enable more patients to achieve and sustain their glycaemic goals with enhanced efficacy and prolonged durability, providing a substantial alternative to the conventional stepwise antihyperglycaemic treatment regimen.[11–13]

Henagliflozin is an orally administered, highly selective SGLT2i with over 800-fold greater selectivity for SGLT2 compared with SGLT1.[14] Two phase 3 clinical trials have shown that henagliflozin, both as monotherapy and as an add-on to metformin, improves glycaemic control, reduces body weight and lowers BP in patients with T2DM.[15,16] In addition, its safety profile aligns with that of other SGLT2is.[17] Retagliptin, a DPP-4i, was approved for T2DM treatment in China on 27 June 2023. In two phase 3 clinical studies (NCT01970033, Juming Lu, Chinese PLA General Hospital, Beijing, China; unpublished data on file with Jiangsu Hengrui Pharmaceuticals; NCT05054842, Lixin Guo, Beijing Hospital, Beijing, China; submitted for publication), retagliptin at 100 mg once daily, either as a monotherapy or as an add-on to metformin, showed clinically significant improvements in glycaemic control among patients with T2DM, with a favourable safety and tolerability profile.

In this trial, we aimed to evaluate whether the co-administration of retagliptin (100 mg) and henagliflozin (5 or 10 mg) once daily, the pioneering DPP-4i and SGLT2i drugs independently developed in China, resulted in superior glycemic control after 24 weeks compared to their individual agents in patients with T2DM who were inadequately controlled with metformin.

2 | MATERIALS AND METHODS

2.1 | Study design

The phase 3 multicentre trial (NCT04667143), conducted across 72 sites in China, featured a randomized, double-blind, active-controlled, parallel-group design over a 24-week treatment period (Figure S1 in Data S1). The study was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines. The trial protocol, along with any amendments, received approval from the independent ethics committees at the participating centres. Before enrolment, all patients provided written, informed consent. The trial sponsor (Jiangsu Hengrui Pharmaceuticals Co., Ltd) designed and oversaw the conduct of the trial. Trial site investigators were responsible for data collection, and the sponsor undertook site monitoring, data collation and data analysis.

2.2 | Patients

Patients were eligible for inclusion if they met the following criteria: (a) aged between 18 and 75 years; (b) diagnosed with T2DM according to the 1999 World Health Organization criteria; (c) taking a stable dose of metformin (≥1500 mg/day) for at least 8 weeks before enrolment; (d) an HbA1c level between 7.5 and 10.5%, and fasting plasma glucose (FPG) of ≤15 mmol/L, confirmed at both screening (by local test) and randomization (by central test); and (e) body mass index >19 and ≤40 kg/m^2. Key exclusion criteria included type 1 or secondary diabetes; acute metabolic complications (e.g. ketoacidosis and lactic acidosis) within the past 6 months; a history of cardiovascular disease (including New York Heart Association class III or IV congestive heart failure, unstable angina, myocardial infarction, severe cardiac arrhythmia, or cardiac surgery) within the past 6 months; a history of acute or chronic pancreatitis or high-risk factors that could potentially lead to pancreatitis; recurrent urinary tract or genital infections; an estimated glomerular filtration rate (eGFR) of <60 ml/min/1.73 m^2; treatment with any anti-hyperglycaemic agents other than metformin within 2 months prior to screening; and systolic BP (SBP) ≥160 mmHg or diastolic BP (DBP) ≥100 mmHg.

2.3 | Randomization and study procedures

Throughout the trial, starting from the lead-in phase, patients were switched to metformin extended-release tablets at the nearest dose (either 1500 or 2000 mg/day), along with lifestyle modifications that included diet and exercise.

After a screening period of <2 weeks, patients received a placebo to assess treatment compliance during a 4-week, single-blind, lead-in period; subsequently, patients with sufficient compliance (between 80% and 120%) who met the glycaemic criteria for inclusion were randomized. Patients were randomly allocated in a 1:1:1:1:1 ratio to one of five treatment groups: retagliptin 100 mg (R100); henagliflozin 5 mg (H5); henagliflozin 10 mg (H10); co-administration of retagliptin 100 mg and henagliflozin 5 mg (R100/H5); or co-administration of retagliptin 100 mg and henagliflozin 10 mg (R100/H10), using a centralized interactive web response system. Randomization was stratified by baseline HbA1c levels (≤8.5% or >8.5%) and the switched dosage of metformin extended-release tablets (either 1500 or 2000 mg/day). An investigator at each site registered patients via the web response system and assigned them on the basis of the randomization sequence directly obtained from the system. Oral retagliptin and henagliflozin were each administered as separate tablets, taken once daily at approximately the same time each morning, regardless of food intake. Both medications were packaged to appear identical to their respective dummy tablets to maintain the integrity of the double-blind study design. From the point of randomization until the database was locked, all participants, including patients, investigators, study site personnel and the sponsor, were blinded to the treatment assignments. Patients were instructed on self-monitoring blood glucose, and those who did not achieve adequate glycaemic control and qualified for rescue therapy were eligible to receive open-label treatment with sulphonylureas or basal insulin, at the investigator's discretion.

2.4 | Endpoints and assessments

The primary endpoint was the change in HbA1c from baseline to week 24, as measured by a central laboratory. Secondary efficacy endpoints included changes from baseline in FPG, 2-h postprandial plasma glucose (2 h PPG), body weight, SBP and DBP at week 24; the proportion of patients achieving HbA1c ≤ 6.5 and <7.0% at week 24; as well as the proportion of patients requiring rescue medicine across all treatment arms. Safety was assessed through a comprehensive evaluation that included monitoring of reported adverse events (AEs), tracking of hypoglycaemic events, measurements of vital signs, physical examinations, analysis of electrocardiographic data and assessment of changes in key clinical laboratory parameters.

Patients who exhibited symptoms of hypoglycaemia or those with recorded blood glucose levels of ≤70 mg/dl (3.9 mmol/L), regardless of symptoms, were instructed to record these events in a hypoglycaemia assessment log (definitions of hypoglycaemia are shown in Table S1 in Data S1). The formula used to calculate patient compliance at each scheduled visit is as follows: [(number of tablets dispensed − number of tablets remaining)/number of tablets prescribed] × 100%.

2.5 | Statistical analysis

The primary study hypotheses posited that, after 24 weeks of treatment, the co-administration groups (R100/H5 or R100/H10), in addition to metformin, would provide a greater reduction in baseline HbA1c than the addition of any individual agent (R100, H5, or H10) at the corresponding dose as an add-on to metformin. We calculated that 121 patients per group would be sufficient for 90% power to detect a 0.5% difference in reduction from baseline HbA1c at week 24 between the R100/H5 group and the H5 group, the R100/H10 group and the H10 group, and either the co-administration group or the R100 group. This was based on a two-sided test with a type I error rate of 0.025 and an assumed standard deviation of 1.1%. Considering a 20% drop-out rate, each group would require 152 patients, resulting in a total sample size of 760.

The Bonferroni correction method is used to control type I error, with a two-sided alpha level set at 0.05. For multiple testing, the R100/H10 group is considered superior to the R100 and H10 groups if the *p*-values from both tests are ≤.025. Similarly, the R100/H5 group is considered superior to the R100 and H5 groups if the *p*-values from both tests are ≤.025. Multiplicity correction is not conducted for secondary efficacy endpoints.

Efficacy was assessed in the full analysis set, which comprised all randomized patients who received at least one dose of the assigned study treatment. The efficacy endpoints of H5/R100 and H10/R100 were compared with those of the equivalent doses of henagliflozin alone and R100 alone. The comparison for the primary endpoint was conducted through two estimands (both primary and supplementary), and the comparison for the secondary efficacy endpoints was conducted only under the primary estimand. For each estimand, two intercurrent events were considered: discontinuation of the study treatment or withdrawal from the study, and the initiation of rescue medication. Study treatment discontinuation or withdrawal from the study was addressed using a hypothetical strategy for both estimands; consequently, data collected after these events were excluded from the efficacy analysis. The initiation of rescue medication was addressed using a treatment policy strategy for the primary estimand and a hypothetical strategy for the supplementary estimand. This approach was taken to assess the treatment effect with or without the potential confounding impact of rescue medication use, respectively. The details on strategies for addressing intercurrent events are listed in Table S2 in Data S1.

Treatment differences for the primary endpoint were performed using an analysis of mixed-effects model repeated measures models based on restricted estimation maximum likelihood methods. The baseline HbA1c value was used as a covariate, while the treatment group, visit, switched metformin extended-release tablets (either 1500 or 2000 mg/day), and the interaction between treatment and visit were considered fixed effects. The patients were treated as random effects, and an unstructured covariance matrix was used. The same mixed-effects model repeated measures model was applied to evaluate changes from baseline in FPG, 2 h PPG, body weight, SBP and DBP at week 24. Least-squares (LS) mean changes from baseline were calculated for each group, accompanied by 95% confidence intervals (CIs). The LS mean differences, along with the corresponding 95% CIs, between the R100/H5 or R100/H10 group and the corresponding doses of individual agents were also provided, accompanied by *p* values for pairwise comparisons.

Prespecified sensitivity analyses for the primary endpoint, based on the primary estimand, were conducted using various models (Tables S3-S5 in Data S1). The proportions of patients achieving HbA1c ≤6.5% or <7.0% at week 24 were analysed with non-responder imputation (patients with missing HbA1c values at week 24 were considered non-responders), and 95% CIs were computed by the Clopper-Pearson method. These proportions were compared between the R100/H5 or the R100/H10 group and the corresponding doses of individual agents using a Cochran-Mantel-Haenszel test, with baseline HbA1c (≤8.5% or >8.5%) and the dosage of switched extended-release metformin tablets (either 1500 or 2000 mg/day) serving as stratification factors. The safety set included all enrolled patients who received at least one dose of study treatment. Statistical analyses were performed using SAS 9.4 software (SAS Institute Inc., Cary, NC, USA).

3 | RESULTS

3.1 | Patient disposition and baseline characteristics

Patients were enrolled from 15 April 2021 to 23 November 2022. All 778 randomized patients received at least one dose of the assigned study treatment, thereby being included in the efficacy and safety analyses; 728 of them (93.6%, ranging from 89.7% to 95.5% across groups) completed the 24-week treatment period (Figure 1). The most common reason for study treatment discontinuation was patient decision (25/778; 3.2%), followed by AEs (10/778; 1.3%).

Baseline demographics and disease characteristics were generally balanced across all study groups (Table 1). The mean age of the patients was 54.6 years, and 64.3% were men. The mean baseline HbA1c level was 8.4%, the mean eGFR was 128.0 ml/min/1.73 m^2, and the median duration of T2DM was 56.2 months. The proportion of patients taking background metformin at a dosage of 1500 and 2000 mg/day was 77.6% and 22.4%, respectively.

3.2 | Efficacy

At week 24, significantly greater reductions in HbA1c from baseline were observed in the co-administration treatment groups (R100/H10 and R100/H5) compared with the groups treated with individual agents at corresponding doses, as per the primary estimand (Table 2 and Figure 2A). The R100/H10 group showed a significant reduction in HbA1c change from baseline compared with both the H10 group [LS mean difference −0.59% (95% CI −0.75 to −0.43); *p* < .0001] and the R100 group [LS mean difference −0.56% (95% CI −0.72 to −0.40); *p* < .0001]. Similarly, the R100/H5 group exhibited a significant reduction in HbA1c change from baseline compared with the H5 group [LS mean difference −0.65% (95% CI −0.81 to −0.49); *p* < .0001] and the R100 group [LS mean difference −0.52% (95% CI −0.68 to −0.36); *p* < .0001]. Consistent reductions in HbA1c from baseline were observed for the supplementary estimand (see Table 2 and Figure 2B). Furthermore, the robustness of the primary endpoint was confirmed through sensitivity analyses (detailed in Tables S3-S5 in Data S1).

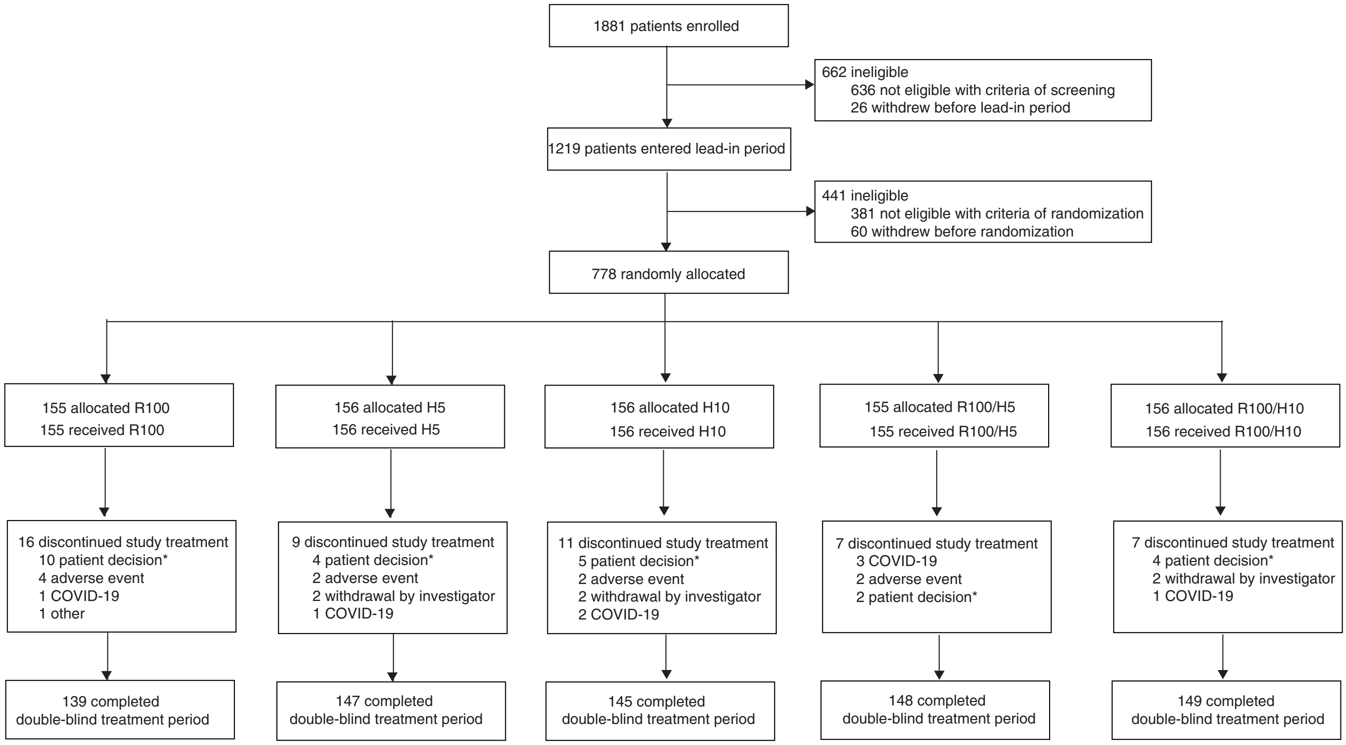

FIGURE 1 Trial profile. COVID-19, coronavirus disease 2019; H10, henagliflozin 10 mg; H5, henagliflozin 5 mg; R100, retagliptin 100 mg. *Regarding the specific reason for discontinuation of study treatment classified under 'patient decision', five patients in the R100 group, one in the H5 group, two in the H10 group, none in the R100/H5 group, and two in the R100/H10 group withdrew from the trial because of lack of efficacy.

At week 24, the proportions of patients achieving HbA1c ≤6.5% were 7.1%, 9.6%, 10.9%, 34.8% and 30.1% in the R100, H5, H100, R100/H5 and R100/H10 groups, respectively. Similarly, HbA1c <7.0% was achieved by 27.1, 21.2, 24.4, 57.4 and 56.4% of patients in these respective groups. Greater proportions of patients achieved HbA1c levels of ≤6.5 and <7.0% at week 24 when treated with the co-administration of retagliptin and henagliflozin, compared with those receiving corresponding doses of the individual agents (all *p*-values <.0001; Table 2 and Figures S2A and S2B in Data S1).

Similarly, improvements in other secondary glycaemic variables at week 24, such as reductions from baseline in FPG (Figure S3A in Data S1) and 2-h PPG (Figure S3B in Data S1), were significantly greater in the groups receiving co-administered retagliptin and henagliflozin compared with those receiving corresponding doses of each individual agent (all *p*-values ≤.003; Table 2). At week 24, reductions in body weight were observed in both the R100/H5 group [LS mean difference −0.99 kg (95% CI −1.51 to −0.45), $p = .0003$] and the R100/H10 group [LS mean difference −2.00 kg (95% CI −2.52 to −1.46), $p < .0001$] when compared with the R100 group (Table 2). In addition, a reduction in SBP was observed in the R100/H10 group at week 24, with an LS mean difference of −3.82 mmHg (95% CI −6.13 to −1.52; $p = .0012$) compared with the R100 group (Table 2). Only one patient (0.6%) in each of the H5, H10, R100/H5 and R100/H10 groups required rescue therapy during weeks 12-24, a rate lower than the 11 patients (7.1%) in the R100 group.

3.3 | Safety

Safety data for the 24-week treatment period are presented in Table 3. Mean treatment compliance was consistently high across all five treatment groups: 99.0% for the R100 group, 98.7% for the H5 group, 98.6% for both the H10 and R100/H5 groups, and 98.5% for the R100/H10 group. Treatment discontinuation because of AEs was low across all study groups: 1.9% in the R100 group, 1.3% in each of the H5, H10 and R100/H5 groups, and no discontinuations in the R100/H10 group (Table S6 in Data S1). Serious AEs were reported as follows: six patients (3.9%) in the R100 group, nine patients (5.8%) in each of the H5, H10 and R100/H5 groups, and three patients (1.9%) in the R100/H10 group. Among these, only two serious AEs in the H10 group (one with diabetic ketosis and another with cerebral ischaemia) were considered definitely or possibly related to the study treatment (Table S7 in Data S1). Both patients achieved a successful recovery following appropriate medical interventions. No deaths were reported during the trial.

AEs that occurred in at least 5% of patients in any treatment group comprised decreased weight, hyperuricaemia, hyperlipidaemia, increased lipase, COVID-19, urine ketone body present, glucose urine present, upper respiratory tract infection, and the urine was positive for white blood cells (Table 3). The frequency of treatment-related AEs showed no meaningful difference across all treatment groups.

TABLE 1 Demographics and baseline characteristics in the full analysis set.

Characteristics	R100 (n = 155)	H5 (n = 156)	H10 (n = 156)	R100/H5 (n = 155)	R100/H10 (n = 156)
Age, years	53.7 ± 8.9	54.8 ± 10.5	54.3 ± 10.2	54.9 ± 10.9	55.2 ± 9.8
Men, n (%)	97 (62.6)	101 (64.7)	109 (69.9)	94 (60.6)	99 (63.5)
Body weight, kg	67.6 ± 11.9	69.1 ± 11.5	70.0 ± 12.7	68.4 ± 12.3	67.9 ± 11.5
BMI, kg/m^2; n (%)	25.0 ± 3.2	25.6 ± 3.2	25.5 ± 3.4	25.4 ± 3.3	25.2 ± 3.2
≥24 kg/m^2	90 (58.1)	102 (65.4)	100 (64.1)	99 (63.9)	105 (67.3)
<24 kg/m^2	65 (41.9)	54 (34.6)	56 (35.9)	56 (36.1)	51 (32.7)
Duration of T2DM, months	70.7 ± 61.5	72.8 ± 67.9	70.3 ± 58.4	69.1 ± 65.2	73.1 ± 59.3
≥56.1 months, n (%)[a]	77 (49.7)	72 (46.2)	85 (54.5)	74 (47.7)	80 (51.3)
<56.1 months, n (%)[a]	77 (49.7)	83 (53.2)	71 (45.5)	81 (52.3)	76 (48.7)
Background metformin, n (%)					
1500 mg/day	121 (78.1)	121 (77.6)	122 (78.2)	120 (77.4)	120 (76.9)
2000 mg/day	34 (21.9)	35 (22.4)	34 (21.8)	35 (22.6)	36 (23.1)
Fasting C-peptide, nmol/L	0.72 ± 0.28	0.74 ± 0.30	0.81 ± 0.35	0.76 ± 0.36	0.72 ± 0.29
HbA1c, %	8.4 ± 0.6	8.4 ± 0.6	8.4 ± 0.7	8.4 ± 0.6	8.5 ± 0.7
HbA1c, n (%)					
>8.5%	59 (38.1)	60 (38.5)	58 (37.2)	61 (39.4)	61 (39.1)
≤8.5%	96 (61.9)	96 (61.5)	98 (62.8)	94 (60.6)	95 (60.9)
FPG, mmol/L	10.7 ± 2.0	10.5 ± 1.9	10.4 ± 2.0	10.5 ± 1.9	10.7 ± 1.8
SBP, mmHg	123.6 ± 12.4	126.9 ± 13.9	124.6 ± 14.2	124.7 ± 12.3	124.4 ± 13.9
DBP, mmHg	80.8 ± 8.9	80.3 ± 8.6	80.5 ± 9.0	80.7 ± 8.2	80.2 ± 8.2
eGFR, ml/min/1.73 m^2	129.3 ± 33.1	125.8 ± 31.8	124.1 ± 32.1	131.5 ± 35.5	129.5 ± 34.2

Note: Continuous data are mean ± standard deviation.
Abbreviations: BMI, body mass index; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; FPG, fasting plasma glucose; H10, henagliflozin 10 mg; H5, henagliflozin 5 mg; HbA1c, glycated haemoglobin; R100, retagliptin 100 mg; SBP, systolic blood pressure; T2DM, type 2 diabetes mellitus.
[a]Median duration of T2DM.

During the double-blind treatment period, 16 patients (2.1%) across all treatment groups experienced prespecified AEs of special interest (Table S8 in Data S1). In the R100, H5, H10, R100/H5 and R100/H10 groups, increased lipase levels were observed in four (2.6%), three (1.9%), four (2.6%), one (0.6%) and four (2.6%) patients, respectively; of these, two patients in the R100 group, three in both the H5 and H10 groups, one in the R100/H5 group and four in the R100/H10 group were deemed treatment-related. In the H10 group, one patient (0.6%) had an acute myocardial infarction, not considered treatment-related per the investigator. Thirty-three patients (4.2%) experienced documented hypoglycaemic events; of these, 23 were classified as Level 1, and 6 as Level 2, with no instances of Level 3 reported (Table S9 in Data S1). No AEs of special interest of acute/chronic pancreatitis, neoplastic events, or episodes of severe hypoglycaemia were reported.

The incidence of urinary tract infections (UTIs) was 1.9%, 2.6%, 2.6%, 4.5% and 1.9% in the R100, H5, H10, R100/H5 and R100/H10 treatment groups, respectively. The incidence of genital infections in groups treated with henagliflozin (1.3% in H5, 0.6% in H10, 1.3% in R100/H5, 1.9% in R100/H10) was low and comparable with that in the retagliptin-only group (0.6% in R100). A few cases of diabetic ketoacidosis were reported, with one in the H10 group and another in the R100/H5 group. In addition, only one patient in the H5 group experienced hypotension throughout the 24-week period (Table 3).

4 | DISCUSSION

In patients with inadequately controlled T2DM on metformin monotherapy, co-administration of retagliptin 100 mg and henagliflozin (5 or 10 mg) resulted in clinically and statistically significant reductions in HbA1c levels at week 24 compared with individual agents at corresponding doses. Supplementary estimand and sensitivity analyses further supported that co-administration groups experienced significantly greater reductions in HbA1c from baseline at week 24, compared with those treated with individual agents at corresponding doses. Concurrently, co-administration of retagliptin and henagliflozin led to synchronized improvements in other glycaemic measures, such as FPG and 2 h PPG. This treatment approach also raised the proportion of patients achieving HbA1c levels of ≤6.5 and <7.0% at week 24. These findings are consistent with those from other phase 3 clinical trials involving the co-administration of different DPP-4is and SGLT2is as add-ons to metformin in similar clinical settings.[11–13]

TABLE 2 Primary endpoint and secondary efficacy endpoints at week 24 in the full analysis set.

Endpoints	R100 (n = 155)	H5 (n = 156)	H10 (n = 156)	R100/H5 (n = 155)	R100/H10 (n = 156)
HbA1c, %; primary estimand					
n[a]	138	145	144	146	148
Change from baseline, LS mean difference (95% CI)	−0.98 (−1.10, −0.87)	−0.86 (−0.97, −0.74)	−0.95 (−1.07, −0.84)	−1.51 (−1.62, −1.39)	−1.54 (−1.66, −1.43)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.65 (−0.81, −0.49); <.0001	-
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−0.59 (−0.75, −0.43); <.0001
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.52 (−0.68, −0.36); <.0001	−0.56 (−0.72, −0.40); <.0001
HbA1c, %; supplementary estimand					
n[a]	128	144	143	145	147
Change from baseline, LS mean difference (95% CI)	−0.93 (−1.05, −0.80)	−0.85 (−0.97, −0.74)	−0.95 (−1.07, −0.83)	−1.50 (−1.62, −1.39)	−1.54 (−1.66, −1.43)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.65 (−0.81, −0.49); <.0001	-
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−0.59 (−0.75, −0.43); <.0001
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.58 (−0.74, −0.41); <.0001	−0.61 (−0.78, −0.45); <.0001
FPG, mmol/L					
n[a]	137	144	142	146	148
Change from baseline, LS mean difference (95% CI)	−1.38 (−1.62, −1.13)	−1.76 (−2.0, −1.5)	−2.19 (−2.43, −1.95)	−2.37 (−2.60, −2.13)	−2.85 (−3.09, −2.62)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.61 (−0.93, −0.28); .0003	-
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−0.66 (−0.99, −0.33); <.0001
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.99 (−1.32, −0.66); <.0001	1.47 (−1.81, −1.14); <.0001
2 h PPG, mmol/L					
n[a]	135	144	141	143	147
Change from baseline, LS mean difference (95% CI)	−4.13 (−4.57, −3.70)	−3.93 (−4.35, −3.51)	−3.97 (−4.40, −3.54)	−5.92 (−6.34, −5.49)	−6.18 (−6.59, −5.76)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	−1.98 (−2.56, −1.41); <.0001	-
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−2.21 (−2.78, −1.64); <.0001
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−1.78 (−2.37, −1.20); <.0001	−2.04 (−2.62, −1.47); <.0001
Body weight, kg					
n[a]	137	144	143	145	148
Change from baseline, LS mean difference (95% CI)	−0.01 (−0.40, 0.38)	−1.76 (−2.13, −1.38)	−1.85 (−2.22, −1.47)	−0.99 (−1.37, −0.62)	−2.00 (−2.37, −1.62)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	0.76 (0.24, 1.29); .0044	-

(Continues)

TABLE 2 (Continued)

Endpoints	R100 (n = 155)	H5 (n = 156)	H10 (n = 156)	R100/H5 (n = 155)	R100/H10 (n = 156)
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−0.15 (−0.68, 0.38); .5729
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−0.98 (−1.51, −0.45); .0003	−1.99 (−2.52, −1.46); <.0001
SBP, mmHg					
n[a]	137	144	143	146	148
Change from baseline, LS mean difference (95% CI)	−0.33 (−2.03, 1.36)	−3.88 (−5.53, −2.22)	−3.11 (−4.77, −1.46)	−2.11 (−3.75, −0.47)	−4.16 (−5.79, −2.53)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	1.77 (−0.51, 4.05); .1287	-
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−1.04 (−3.32, 1.23); .3685
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−1.78 (−4.08, 0.53); .1315	−3.82 (−6.13, −1.52); .0012
DBP, mmHg					
n[a]	137	144	143	146	148
Change from baseline, LS mean difference (95% CI)	−0.50 (−1.63, 0.62)	−2.09 (−3.19, −1.00)	−1.41 (−2.51, −0.31)	−1.86 (−2.95, −0.77)	−1.48 (−2.56, −0.40)
Change versus H5, LS mean difference (95% CI); *p*-value[b]	-	-	-	0.23 (−1.28, 1.74); .7642	-
Change versus H10, LS mean difference (95% CI); *p*-value[b]	-	-	-	-	−0.07 (−1.58, 1.44); .9242
Change versus R100, LS mean difference (95% CI); *p*-value[b]	-	-	-	−1.36 (−2.88, 0.17); .0821	−0.98 (−2.50, 0.55); .2087
Patients with HbA1c ≤6.5%					
n/N[c]	11/155	15/156	17/156	54/155	47/156
Proportion of patients, % (95% CI)[d]	7.1 (3.6, 12.3)	9.6 (5.5, 15.4)	10.9 (6.5, 16.9)	34.8 (27.4, 42.9)	30.1 (23.1, 38.0)
Difference versus H5 (95% CI)[e]	-	-	-	25.4 (16.7, 34.1)	-
Odds ratio versus H5 (95% CI); *p*-value[f]	-	-	-	5.4 (2.8, 11.3); <.0001	-
Difference versus H10 (95% CI)[e]	-	-	-	-	19.5 (11.0, 28.0)
Odds ratio versus H10 (95% CI); *p*-value[f]	-	-	-	-	3.9 (2.0, 7.5); <.0001
Difference versus R100 (95% CI)[e]	-	-	-	28.0 (19.7, 36.3)	23.2 (15.0, 31.3)
Odds ratio versus R100 (95% CI); *p*-value[f]	-	-	-	8.2 (3.8, 18.1); <.0001	6.1 (2.8, 13.4); <.0001
Patients with HbA1c <7.0%					
n/N[c]	42/155	33/156	38/156	89/155	88/156
Proportion of patients, % (95% CI)[d]	27.1 (20.3, 34.8)	21.2 (15.0, 28.4)	24.4 (17.9, 31.9)	57.4 (49.2, 65.3)	56.4 (48.2, 64.3)
Difference versus H5 (95% CI)[e]	-	-	-	36.5 (26.8, 46.3)	-
Odds ratio versus H5 (95% CI); *p*-value[f]	-	-	-	5.8 (3.4, 10.5); <.0001	-
Difference versus H10 (95% CI)[e]	-	-	-	-	32.6 (22.7, 42.5)

TABLE 2 (Continued)

Endpoints	R100 (n = 155)	H5 (n = 156)	H10 (n = 156)	R100/H5 (n = 155)	R100/H10 (n = 156)
Odds ratio versus H10 (95% CI); *p*-value[f]	-	-	-	-	4.7 (2.7, 8.1); <.0001
Difference versus R100 (95% CI)[e]	-	-	-	30.7 (20.6, 40.8)	29.7 (19.6, 39.8)
Odds ratio versus R100 (95% CI); *p*-value[f]	-	-	-	4.1 (2.4, 7.0); <.0001	3.9 (2.3, 6.7); <.0001

Abbreviations: CI, confidence interval; DBP, diastolic blood pressure; FPG, fasting plasma glucose; HbA1c, glycated haemoglobin; LS, least squares; 2 h PPG, 2 h postprandial plasma glucose; SBP, systolic blood pressure.

[a]Number of patients with observed values at week 24.

[b]R100/H5 or R100/H10 were compared with the corresponding doses of individual agents using a mixed-effects model for repeated measures based on the restricted maximum likelihood method. Baseline HbA1c value served as a covariate, while the treatment group, visit, dosage of switched extended-release metformin tablets (either 1500 or 2000 mg/day), and the interaction between treatment and visit were considered fixed effects. Patients were treated as random effects, and an unstructured covariance matrix was employed.

[c]Number of patients achieving HbA1c ≤6.5% or HbA1c <7.0% at week 24/number of patients with observed values analysed with non-responder imputation (patients with missing HbA1c value at week 24 were considered non-responders) at week 24.

[d]95% CIs were calculated using the Clopper-Pearson method.

[e]Proportion differences and corresponding 95% CIs were calculated based on the Mantel-Haenszel estimate.

[f]Proportions of achieving HbA1c ≤6.5% or HbA1c <7.0% at week 24 were compared between the R100/H5 or the R100/H10 group and the corresponding doses of individual agents using Cochran-Mantel-Haenszel tests, stratified by baseline HbA1c levels (≤8.5% or >8.5%) and the dosage of their background metformin extended-release tablets (either 1500 or 2000 mg/day), and the corresponding odds ratio and 95% CI values were calculated.

In our trial, treatment groups containing henagliflozin led to a noticeable reduction in SBP of −2.11 to −4.16 mmHg at week 24 when henagliflozin was administered either as an individual agent or in combination with retagliptin, in contrast to a negligible reduction of −0.33 mmHg in patients receiving retagliptin as an individual agent. While the exact antihypertensive mechanisms of SGLT2is have not been fully elucidated, potential pathways are thought to include increased urinary sodium excretion and enhancements in vascular endothelial function.[18,19] Emerging evidence from pivotal clinical trials (EMPA-REG OUTCOME, CANVAS Program, DECLARE-TIMI 58) suggests that SGLT2is may be efficacious in preventing heart failure among patients with T2DM.[7,9,20,21] This preventative effect could be partially attributed to the reduction in SBP observed in regimens that include henagliflozin.

Adding henagliflozin to metformin, either alone or in combination with retagliptin, resulted in a clinically meaningful change from baseline in body weight at week 24: −1.76 kg for the H5 group, −1.85 kg for the H10 group, −0.99 kg for the R100/H5 group, and −2.00 kg for the R100/H10 group. These results contrasted with the minimal change of −0.01 kg observed in patients treated solely with retagliptin (R100 group). Except for the R100/H5 group, the trends in body weight reduction were consistent with findings from studies on the co-administration of other SGLT2is and DPP-4is.[11–13] In our trial, the magnitude of body weight reduction observed in groups receiving henagliflozin was numerically lower than that reported in two other phase 3 trials involving henagliflozin among patients with T2DM.[15,16] This difference was particularly notable in the modest reductions in body weight and SBP seen in the R100/H5 group. Such discrepancies in body weight and SBP reductions, specifically between the H5 and R100/H5 groups, are unexpected, as previous studies had not reported that the efficacy of SGLT2is in reducing body weight and SBP would be diminished when co-administered with a DPP-4i. Given that the effects of SGLT2is on BP and body weight are relatively mild compared with their robust glucose-lowering effect, they are more prone to confounding factors that could introduce biases. The differential impacts of COVID-19 pandemic lockdowns on study patients could be a primary confounding factor contributing to the observed incongruences. Variations in lockdown durations, policies and resource availability across different regions could have influenced patients' dietary and exercise habits, introducing a bias in the results related to body weight and SBP changes. Furthermore, the study design did not incorporate site as a stratified random factor, suggesting that this bias could not be fully addressed through randomization.

The co-administration of retagliptin and henagliflozin over a 24-week period was safe, with high adherence and no new safety concerns, aligning with the AEs expected from DPP-4is and SGLT2is.[22–24] The incidences of UTIs were similar across treatment groups, supported by a meta-analysis of 86 trials showing no increased UTI risk with SGLT2is compared with placebo or other active treatments.[25] Incidences of hypotension and diabetic ketoacidosis were rare. Furthermore, there was no clinically meaningful difference in the occurrence rate of documented hypoglycaemic events between the groups receiving the co-administration of retagliptin and henagliflozin versus

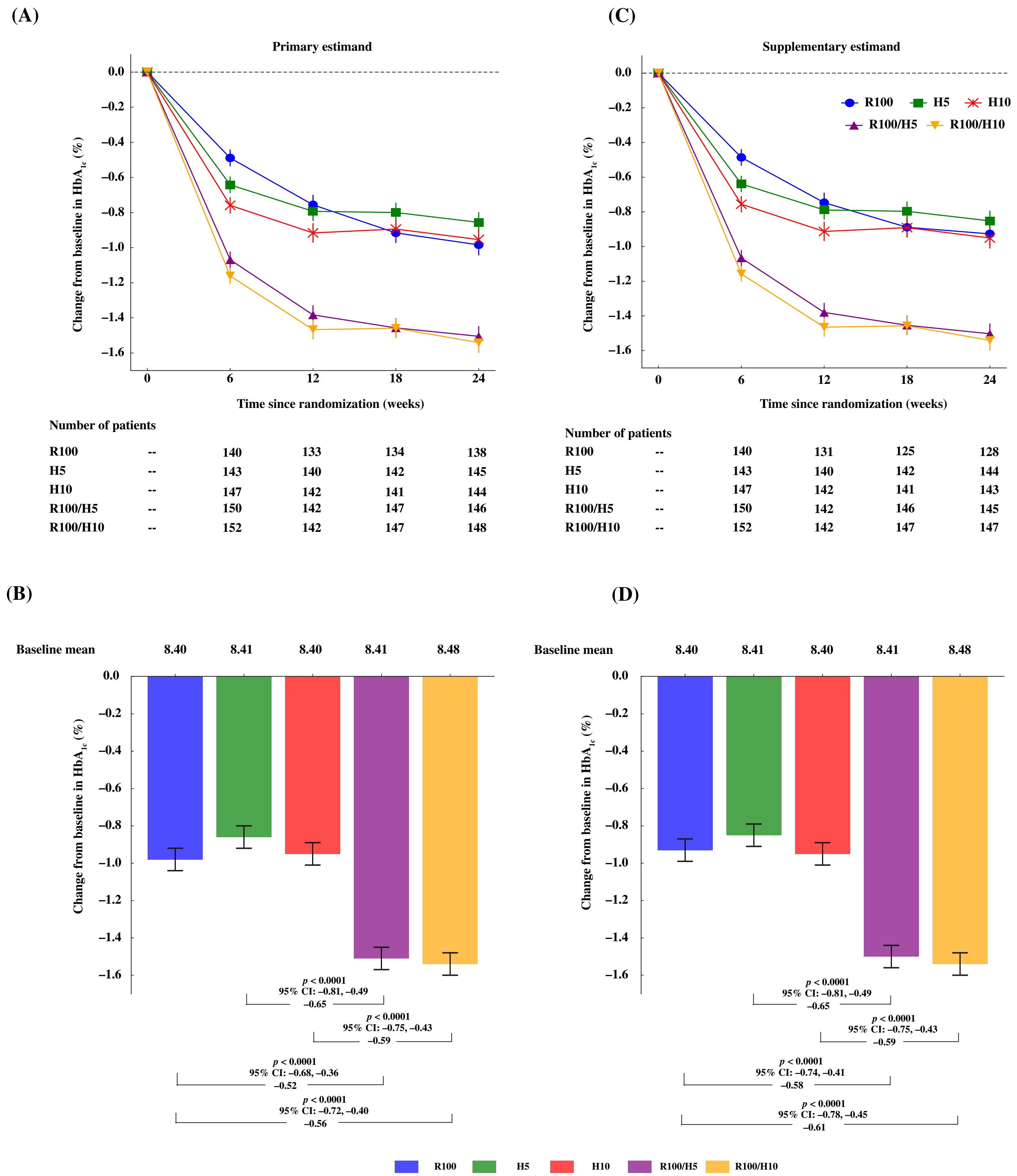

FIGURE 2 Change from baseline in HbA1c. (A) Primary estimand: change from baseline in HbA1c over 24 weeks. (B) Supplementary estimand: change from baseline in HbA1c over 24 weeks. (C) Primary endpoint (primary estimand): change from baseline in HbA1c at week 24. (D) Primary endpoint (supplementary estimand): change from baseline in HbA1c at week 24. Data are least squares mean ± SE. H10, henagliflozin 10 mg; H5, henagliflozin 5 mg; HbA1c, glycated haemoglobin; R100, retagliptin 100 mg; SE, standard error. Upper row: blue circles (R100 group), green squares (H5 group), red crosses (H10 group), purple triangles (R100/H5 group), yellow inverted triangles (R100/H10 group). Lower row: bar chart color coding matches the upper row scheme.

TABLE 3 Adverse events during the 24-week treatment period in the safety set.

AEs	R100 (n = 155)	H5 (n = 156)	H10 (n = 156)	R100/H5 (n = 155)	R100/H10 (n = 156)
AEs ≥1	121 (78.1)	118 (75.6)	119 (76.3)	114 (73.5)	107 (68.6)
AEs related to study drug[a]	56 (36.1)	63 (40.4)	57 (36.5)	59 (38.1)	57 (36.5)
Serious AEs	6 (3.9)	9 (5.8)	9 (5.8)	9 (5.8)	3 (1.9)
Deaths	0	0	0	0	0
Discontinuations following an AEs	3 (1.9)	2 (1.3)	2 (1.3)	2 (1.3)	0
Prespecified AESIs[b]	4 (2.6)	3 (1.9)	4 (2.6)	1 (0.6)	4 (2.6)
Pancreatitis-associated events[c]	4 (2.6)	3 (1.9)	3 (1.9)	1 (0.6)	4 (2.6)
Acute and/or chronic pancreatitis	0	0	0	0	0
Amylase/lipase >3× ULN	4 (2.6)	3 (1.9)	3 (1.9)	1 (0.6)	4 (2.6)
Neoplastic events[d]	0	0	0	0	0
Major adverse cardiovascular events[e]	0	0	1 (0.6)	0	0
Episodes of severe hypoglycaemia[f]	0	0	0	0	0
Urinary tract infection	3 (1.9)	4 (2.6)	4 (2.6)	7 (4.5)	3 (1.9)
Other events of special interest					
Genital infection[g]	1 (0.6)	2 (1.3)	1 (0.6)	2 (1.3)	3 (1.9)
Diabetic ketoacidosis	0	0	1 (0.6)	1 (0.6)	0
Hypotension	0	1 (0.6)	0	0	0
AEs occurring in ≥5% patients in either group					
Weight decreased	4 (2.6)	8 (5.1)	12 (7.7)	14 (9.0)	22 (14.1)
Hyperuricaemia	23 (14.8)	10 (6.4)	7 (4.5)	7 (4.5)	15 (9.6)
Hyperlipidaemia	20 (12.9)	11 (7.1)	15 (9.6)	16 (10.3)	13 (8.3)
Lipase increased	15 (9.7)	8 (5.1)	10 (6.4)	14 (9.0)	11 (7.1)
COVID-19	9 (5.8)	15 (9.6)	8 (5.1)	7 (4.5)	10 (6.4)
Urine ketone body present	4 (2.6)	10 (6.4)	13 (8.3)	7 (4.5)	7 (4.5)
Glucose urine present	4 (2.6)	10 (6.4)	3 (1.9)	10 (6.5)	7 (4.5)
Upper respiratory tract infection	13 (8.4)	6 (3.8)	11 (7.1)	10 (6.5)	7 (4.5)
White blood cells urine positive	9 (5.8)	5 (3.2)	5 (3.2)	6 (3.9)	5 (3.2)

Note: Data are n (%).

Abbreviations: AE, adverse event; AESI, adverse events of special interest; COVID-19, coronavirus disease 2019; H10, henagliflozin 10 mg; H5, henagliflozin 5 mg; R100, retagliptin 100 mg; ULN, upper limit of normal.

[a]Determined by the investigator to be related to the study drug.

[b]Prespecified AESIs were also recorded and included pancreatitis-associated events (acute and chronic pancreatitis; elevations in amylase or lipase levels >3 times the ULN), neoplastic events, major adverse cardiovascular events and episodes of severe hypoglycaemia.

[c]Acute and chronic pancreatitis observed post-administration. For confirmed cases of pancreatitis, follow-up evaluations and investigations into other potential aetiological factors are recommended. Suggested diagnostic tests include gallbladder ultrasound, lipid profile assays, and any other examinations deemed necessary by the investigator. A comprehensive medical history should also be collected, detailing concurrent medication usage, dietary habits and alcohol consumption. Post-administration elevations in serum amylase or lipase levels >3× ULN are also included.

[d]Occurrence of all categories of malignant tumours following the administration of the study medication.

[e]Major adverse cardiovascular events included cardiovascular-related death, non-fatal myocardial infarctions, hospitalizations because of unstable angina and hospitalizations resulting from heart failure.

[f]Severe hypoglycaemic events were defined as those occurring after medication administration and meeting the Level 3 hypoglycaemia criteria as outlined by the American Diabetes Association's (ADA) 2020 guidelines.

[g]Clustered Medical Dictionary for Regulatory Activities version 25.0 preferred term: vulvovaginal mycotic infection, vaginal infection, vulvitis, vulvovaginal candidiasis, reproductive tract infections, genital candidiasis, genital infection fungal, vulvovaginitis, cervicitis, bacterial vaginosis, vulvovaginal pruritus, genitourinary tract infection, scrotal abscess, penile infection, balanoposthitis, balanitis candida, genital infection, vulvovaginitis trichomonal, pruritus genital, atrophic vulvovaginitis, papilloma viral infection.

those treated with individual agents, which was consistent with previous studies suggesting that the concomitant use of metformin with DPP-4i and/or SGLT2i does not result in an elevated risk of hypoglycaemia.[11–13]

Despite SGLT2is increasing renal glucose excretion,[26] groups containing henagliflozin did not show a higher rate of genital infections compared with those treated with retagliptin as an individual agent. The low incidence of genital infections was consistent with previous phase 3 trials of henagliflozin in patients with T2DM,[15,16] possibly because of rigorous adherence to good hygiene practices and the exclusion of patients with a history of recurrent urinary tract or genital infections. In addition, considering the notably lower incidence of genital mycotic infections in men than in women as documented in SGLT2i studies,[27] the lower incidence of genital infections observed in our study may also be partly attributed to its higher male representation of 64.3%, versus the 50.2%-53.9% observed in similar trials.[11–13]

The current trial has several limitations. First, our study's inclusion criteria were restricted to patients with eGFR >60 ml/min/1.73 m^2 and a minimum daily metformin dose of 1500 mg, aiming to evaluate precisely the efficacy of the trial agents at their maximum recommended dosages. However, many patients with eGFR between 45 and 60 ml/min/1.73 m^2 could benefit from this triple therapy by reducing the dosages of metformin and retagliptin. Moreover, it is common in clinical practice to encounter patients on a daily metformin dose of <1500 mg. In the context of early combination therapy, where multiple drugs exert a hypoglycaemic effect, it is worth discussing both the necessity and the potential additional benefits of using metformin at its maximum dose. Consequently, future studies in real-world settings are necessary to evaluate the benefits of triple therapy in these patient populations. Second, the 24-week duration limited the generalizability of the findings concerning the long-term durability of these benefits and their potential impact on reducing cardiovascular events.

In conclusion, in patients with T2DM that were inadequately controlled by metformin, the co-administration of retagliptin and henagliflozin showed superior glucose-lowering efficacy compared with the individual agents, with a minimal risk of hypoglycaemia. Both retagliptin and henagliflozin were well-tolerated, exhibiting safety profiles consistent with the expected class effects.

AUTHOR CONTRIBUTIONS

Yao Wang, Zi Ye, Danli Zhang, Quanren Wang and Wenying Yang: design. Yao Wang, Jing Xu, Xiaolin Dong, Mingwei Chen, Qin Gu, Lihui Zhang, Yanqin Fu, Tianrong Pan, Yan Bi, Weihong Song, Jing Xu, WeiPing Lu, Xiaodong Sun, Zi Ye, Danli Zhang, Liang Peng, Quanren Wang and Wenying Yang: conduct/data collection. Wei Dai and Xiang Lin: statistical analysis. Yao Wang, Liang Peng, Wei Dai and Wenying Yang: drafting of the manuscript. All authors: interpretation of data, critical revision of the manuscript for important intellectual content and approval of the final manuscript.

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CONFLICT OF INTEREST STATEMENT

ZY, DZ, LP, XL, WD and QW are employees of Jiangsu Hengrui Pharmaceuticals Co., Ltd. All other authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

Data are available upon reasonable request. The data sets used and/or analysed during the current study are available from the corresponding author on reasonable request after the product and indication has been approved by major health authorities. Data may be requested 24 months after study completion. Qualified researchers should submit a proposal to the corresponding author outlining the reasons for requiring the data. The leading clinical site and sponsor will check whether the request is subject to any intellectual property restriction. Use of data must also comply with the requirements of Human Genetics Resources Administration of China and other country or region-specific regulations. A signed data access agreement with the sponsor is required before accessing shared data.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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